

firm regardless of AI exposure. For workers aged 22-25, we find a 15 log-point decline in relative employment for the most AI-exposed quintiles compared to the least exposed quintile, a large and statistically significant effect. Estimates for other age groups are much smaller in magnitude and not statistically significant. These findings imply that the employment trends we observe are not driven by differential shocks to firms that employ a disproportionate share of AI-exposed young workers.

Fifth, the labor market adjustments are visible in employment more than in compensation. In contrast to our findings for employment, we find little difference in annual salary trends by age or exposure quintile, suggesting possible wage stickiness. If so, AI may have larger effects on employment than on wages, at least initially, or even that AI may boost wages for as many workers as it hurts.

Sixth, the above facts are largely consistent across sample constructions designed to address various alternative explanations for the core findings. Our results are not driven solely by computer occupations or by occupations susceptible to remote work and outsourcing. We also find that the AI exposure taxonomy did not meaningfully predict employment outcomes for young workers further back in time, before the widespread use of LLMs, including during the unemployment spike driven by the COVID-19 pandemic. The patterns we observe in the data appear most acutely starting in late 2022 and early 2023, around the time of rapid proliferation of generative AI tools.³ They hold for both occupations with a high share of college graduates and ones with a low college share, suggesting deteriorating education outcomes during COVID-19 do not drive our results. For non-college workers, we find evidence that experience may serve as less of a buffer to labor market disruption, as low college share occupations exhibit divergent employment outcomes by AI exposure up to age 40.

While we explore a variety of alternative explanations, we caution that the facts we document may in part be influenced by factors other than generative AI. Taken as a whole, our results are consistent with the hypothesis that generative AI has begun to affect entry-level employment. We intend to continue to track the data on an ongoing basis to assess whether these trends change in the future.

³In particular, OpenAI introduced ChatGPT in November 2022. It was reported to have over 100 million active users by January 2023 and 1.7 billion website visitors in October 2023 (DeVon, 2023).

4 Results

4.1 Fact 1: Employment for young workers has declined in AI-exposed occupations

Consider software engineers and customer service agents, two occupations frequently considered to be highly exposed to generative AI tools. Media attention has raised the specter of widespread employment disruption for young software engineers (Thompson, 2025; Raman, 2025; Allen, 2025; Horowitch, 2025).

Figure 1 shows employment changes by age group for these occupations, normalized to 1 in October 2022. While the types of work and workers in each of these occupations differs in many ways, both occupations present a similar pattern: employment for the youngest workers declines considerably after 2022, while employment for other age groups continues to grow. By September 2025, employment for software developers aged 22-25 declined nearly 20% compared to its peak in late 2022. Figure A1 shows that a similar pattern holds for computer occupations and service clerks more generally.

Figure A2 shows four other professions as case studies, spanning varying levels of AI exposure according to Eloundou et al. (2024). Marketing and sales managers, in the fourth quintile of AI exposure, show a decline in employment for young workers much like the case of software and customer service, albeit with smaller magnitudes. Front-line production and operations supervisors, in quintile 3, show an increase in employment for young workers, though the growth in employment is smaller than the increase for workers over age 35.

In contrast, the trends for occupations that Eloundou et al. (2024) rated as less exposed do not fit the pattern of the more exposed occupations. Stock clerks and order fillers, in quintile 2, show no obvious difference by age. Strikingly, the series for health aides, comprising nursing aides, psychiatric aides, and home health aides, show a quite different trend from software or customer service: employment for young workers has been growing *faster* than for older workers.

Figure 2 shows these patterns hold more generally across professions. The top left plot shows a divergence in employment outcomes for more and less exposed occupations for workers aged 22-25, with more exposed occupations experiencing declining employment. For older age groups, we find

f indexes firms, q indexes Eloundou et al. (2024) exposure quintiles, and t indexes months, with $t = -1$ corresponding to October 2022. The outcome variable $y_{f,q,t}$ is employment in f, q, t . Equation 4.1 is a Poisson event study regression controlling for firm-quintile effects, $\alpha_{f,q}$, and firm-time effects, $\beta_{f,t}$. The firm-time effects absorb aggregate firm shocks that impact each exposure quintile equally. The firm-quintile effects adjust for baseline differences in hiring across quintiles within the firm. The coefficients of interest, $\gamma_{q,t}$, measure differential changes in employment growth across quintiles after accounting for firm-time effects and firm-quintile effects.¹⁸

We run this regression separately for each age group. For each regression, we restrict to firms that hire at least 10 workers within the age group in every period of the sample. Further, $\sum_t y_{f,q,t}$ must equal at least 100 for each q , meaning that the firm must at least employ on average about 2 workers from each exposure quintile across months in the sample.¹⁹ Standard errors are clustered by firm.

Results are in Figure 4, which plots the $\gamma_{q,t}$ coefficients for each age group. For workers aged 22-25, estimates for higher quintiles are large and statistically significant, with a 15 log point decline in relative employment comparable in magnitude to the estimates in the raw data in Figure 2. Estimates for other age groups are generally much smaller in magnitude and not statistically significant. These findings imply that the employment trends we observe are not driven by differential shocks to firms that employ a disproportionate share of AI-exposed young workers.

One alternative confounder that would not be controlled for with firm-time effects is that even conditional on the firm workers with high AI exposure were excessively hired after the COVID-19 pandemic, leading to a subsequent contraction in their hiring. To assess such alternatives we consider various other robustness checks in Section 4.6, such as removing computer occupations and conditioning on whether the occupation is amenable to work from home.

¹⁸Because of zero counts in the outcome variable, we estimate a Poisson regression instead of an OLS regression in logs following guidance from Chen and Roth (2024).

¹⁹Results are not sensitive to these restrictions, though there must be at least one non-zero value in each firm-month and each firm-quintile for observations to not get dropped in the Poisson regression.